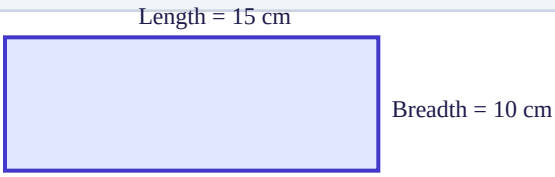
NCERT Mathematics — Class 6

Chapter 6: Perimeter and Area

6.1 Introduction to Perimeter

Look at the figures around you: a picture frame, a rectangular table-top, or a football ground. When we walk around the boundary of a playground, the total distance covered is called its **perimeter**.

Definition: Perimeter is the total distance covered along the boundary forming a closed figure when you go around the figure once.



Perimeter of a Rectangle:
Perimeter = Length + Breadth + Length + Breadth = 2 × (Length + Breadth)

Perimeter of a Square:
Perimeter = 4 × Side

Perimeter of an Equilateral Triangle:
Perimeter = 3 × Side

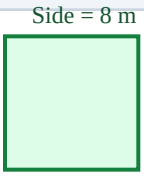
Example 1: Shabana wants to put a lace border all around a rectangular table cover 3 m long and 2 m wide. Find the length of the lace required by Shabana.

Solution:
Length of rectangular table cover (l) = 3 m
Breadth of table cover (b) = 2 m
Length of lace required = Perimeter of table cover
= 2 × (l + b) = 2 × (3 m + 2 m) = 2 × 5 m = 10 m.
Therefore, 10 meters of lace is required.

6.2 Concept of Area

Look at closed figures like squares, triangles and rectangles drawn on paper. The amount of surface enclosed by a closed figure is called its **area**.

Formulas:
• Area of a Rectangle = Length × Breadth (sq units)
• Area of a Square = Side × Side (sq units)



Example 2: Find the area of a square plot of side 8 m.

Solution:
Side of the square plot = 8 m
Area of the square = Side × Side = 8 m × 8 m = 64 sq m (or 64 m^2).

Example 3 (Tiling Word Problem): The floor of a room is 4 m long and 3 m wide. How many square tiles of side 50 cm are required to cover the floor?

Solution:
Length of floor = 4 m = 400 cm
Breadth of floor = 3 m = 300 cm
Area of floor = 400 cm × 300 cm = 1,20,000 cm^2.
Side of one tile = 50 cm
Area of one tile = 50 cm × 50 cm = 2,500 cm^2.
Number of tiles = (Area of floor) ÷ (Area of one tile) = 1,20,000 ÷ 2,500 = 48 tiles.

6.3 Exercise Questions (Practice)

Q.No	Question Description	Marks
1	Find the perimeter of a rectangle whose length is 40 cm and breadth is 10 cm.	1
2	A square piece of land has side 250 m. Find the cost of fencing it at Rs 20 per meter.	2
3	The area of a rectangular garden 50 m long is 300 sq m. Find its width.	2
4	Which has greater area: a square of side 10 cm or a rectangle of 12 cm by 8 cm?	3